

# Milnor number, and Versal Deformation

Convention: by "singularity" we mean:

(a germ of) a holomorphic function  $f: (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$

we care about degeneracies of its critical points  $\{p \mid \frac{\partial f}{\partial x_i}(p) = 0\}$

Defn. A critical point  $p$  is said to be nondegenerate (Morse), if its Hessian matrix  $\left( \frac{\partial^2 f}{\partial x_i \partial x_j} \right) \Big|_{x=p}$  is nondegenerate.

rmk. at a critical point, we get a congruent matrix after change of coordinates.  $J^T A J$   
• nondegenerate  $\Rightarrow$  isolated.

Prop. generic functions are nondegenerate. (in the sense of  $\mathbb{C}^2$  topology)

prop. a nondegenerate function has a "normal form" near 0.  
$$f = x_1^2 + \dots + x_n^2 \quad (\text{after choosing suitable coordinate})$$

For degenerate critical points, we get:

• number :  $\text{corank} = n - \text{rank} \left( \frac{\partial^2 f}{\partial x_i \partial x_j} \right) \Big|_{x=p}$

• algebra: local algebra  $= \mathcal{Q}_f = \mathcal{O}_n / I_f$

where:  $\mathcal{O}_n$ : algebra of function germs at  $0 \in \mathbb{C}^n$

$I_f$  = ideal generated by  $\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}$

defn:  $\mu(f) = \dim_{\mathbb{C}} \mathcal{Q}_f$  is called the multiplicity  
(milnor number)

• space: deformation spaces. (we will come to this later).

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1. Local algebra. (Milnor ring)

Eg. let  $f(z) = z^3$  ( $n=1$ ) then  $I_{\nabla f} = \mathcal{M}^2$

$\dim_{\mathbb{C}} \mathcal{Q}_f = 2$ , is generated by 1 and  $z$

Eg. let  $f(x, y) = x^2 y^2$

$\dim_{\mathbb{C}} \mathcal{Q}_f = \infty$   $(xy)^n$  ( $n=1, 2, \dots$ ) are linearly independent  
in  $\mathcal{Q}_f$

Prop (holomorphic case)  $\mu(f) < \infty \iff 0$  is isolated.

Idea:  $I_{\nabla f}$  is the ideal of critical point

proof: from (Arnold, I, Section 5)

$\Rightarrow$ ) any  $1 \leq i \leq n$

1.  $z_i, z_i^2, \dots, z_i^m$  are linearly dependant in  $\mathcal{Q}_f$

$\Rightarrow c_0 + c_1 z_1 + \dots + c_r z_i^m \in I_f$

( $c_0 \neq 0 \Rightarrow c_0 + c_1 z_1 + \dots + c_r z_i^m$  is invertible  $\Rightarrow I_f = \mathcal{O}_n$ )  
 $\Rightarrow 0$  is not a critical point.

$$\Rightarrow z_i^{r_i} (c_r + \dots) \in I_f$$

$$\Rightarrow z_i^{r_i} \in I_f \Rightarrow z(f) \in \bigcap z(z_i^{r_i}) = 0$$

( $\Leftarrow$  By local variant of Hilbert's zero theorem

$$\exists N, \quad x_j^N \in I_{f,0} \quad \#$$

We will mostly focus on isolated singularities. In this case, the milnor number also admit lots of geometric / topological meanings.

Fact (Milnor)  $\mu$  is equal to the mapping degree of the map

$$\frac{\nabla f}{\|\nabla f\|} : S^{\frac{2n-1}{s}} \rightarrow S^{2n-1}, \quad s \ll 1.$$

*Idea of Proof (Arnold)*

$$\nabla f \simeq \nabla \Phi + \varepsilon f \quad \text{with } \Phi = (x_1^{m_1}, \dots, x_n^{m_n}).$$

*index = multiplicity for such maps.*

The mapping degree point-of-view can also be understood through perturbation.

Recall that generic functions are Morse functions.

defn. A morsification of  $f$  is a Morse function  $f_\epsilon$  of form  $f + \epsilon h$  with  $\epsilon$  arbitrary small.

prop.  $\mu$  is the number of critical points of any morsification of  $f$  in the domain  $B_S = \{z \mid \|z\| \leq S\}$

Proof. The mapping degree is homotopic invariant.

$$\deg f = \deg f_\epsilon \quad f \sim f_\epsilon$$

Degree of  $f_\epsilon$  is sum of indices at each critical point.

At each critical point, we get 1 (use normal form, so we get 1 or -1. but holomorphic function keep orientation).

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Another way to think about  $\mu$  is through milnor fibres.

Theorem (Milnor) A nonsingular level manifold

$V_t = f^{-1}(t) \cap U$ , is homotopy equivalent to a wedge of  $\mu$   $(n-1)$ -dimensional spheres.

cor  $\mu$  is the middle betti number of a generic fibre.

rmk  $f^{-1}(a) \simeq f_2^{-1}(a)$  for  $a$  regular value for both...

rmk A generic fibre of a mors function

$f = z_1^2 + \dots + z_n^2$  is diffeomorphic to the tangent bundle of  $S^{n-1}$ .

Proof we consider  $f^{-1}(1)$

$$\text{write } z = x + iy \Rightarrow \begin{cases} x^2 = 1 + y^2 \\ x \cdot y = 0 \end{cases}$$

$$\therefore f^{-1}(1) \cap B(r) = \{ x + iy \mid \begin{matrix} x^2 = 1 + y^2 \\ xy = 0 \end{matrix} \} \\ (x, 0)$$

$\uparrow$

$\cup$

$x$

$$S^{n-1} = \{ x \mid x^2 = 1 \}$$

we finish with observation that

$$y \in T_x S^{n-1} \Leftrightarrow x \cdot y = 0 \quad \#$$

rmk a standard picture that we will use in the future is that, we move to a general morsification  $f_\epsilon$ .

that also separate the critical values.

then a nonsingular level set of  $f_\epsilon$  looks a wedge of  $n$   $(n-1)$ -dimensional spheres.

## versal deformation

Defn A class of singularities is any subset of the space  $\mathcal{O}_n$  of function germs that is invariant under the group of germs of diffeomorphisms  $(\mathbb{C}^n, 0) \rightarrow (\mathbb{C}^n, 0)$ .

Observation, the tangent space to the orbit of a germ  $f$  is generated by  $\left\{ \frac{\partial f}{\partial x_i} \right\}$

Proof any one-parameter families  $g(x, \tau)$  of diffeomorphisms:  $g(x, 0) \equiv x$ .

$$\frac{\partial}{\partial \tau} f(g(x, \tau)) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{dg_i}{d\tau} \bigg|_{\tau=0}$$

functions close to  $f$  can be tested by their deformation.

Defn A deformation (unfolding) of  $f$  with base

$\Lambda = \mathbb{C}^l$  is a function germ  $\bar{F}: (\mathbb{C}^n \times \mathbb{C}^l, 0) \rightarrow \mathbb{C}$ .

such that  $F(x, 0) \equiv f(x)$ .

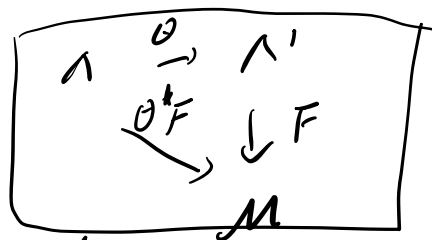
Defn A deformation is said to be

versal, if every deformation of  $f$

is equivalent to one induced from  $\bar{F}$ .

$F(A)$   
 $f = F(0)$

in the sense of pull back:



A versal deformation for which the base has the smallest possible dimension is called miniversal.

Versality can be understood through transversality: in finite-dimensional setting:

Let  $G$  be a Lie group acting on a manifold  $M$ .  
and  $f \in M$

Observation:  $F: \Lambda \rightarrow M$  is versal  $\Rightarrow F$  is transversal to the orbit  $Gf$  of  $f$ . i.e:

$$F_* \overset{\uparrow}{T_0 \Lambda} + T_f Gf = T_f M$$

(differential)

proof. versal  $\Leftrightarrow$  every deformation  $F'$  is pull back  $\theta^* F = F \circ \theta$  with  $\theta: \Lambda' \rightarrow \Lambda$

$$(F')_* (T_0 \Lambda') \subseteq F_* (T_0 \Lambda)$$

we can always find an one parameter family

$F'(\lambda)$  such that  $F'_* \left( \frac{\partial}{\partial \lambda} \right)$  is any given vector in  $T_f M \neq$

Fact (still in finite dimensional setting)

Theorem (Arnold) Any transversal (of minimal dimension) to the orbit of  $f$  is a versal (resp. miniversal) deformation.

— so, how to find a miniversal deformation:

answer to this: find functions  $\varphi_j$  ( $1 \dots m$ ) that form a basis of  $\mathcal{Q}_f$ :

$$F(x, \lambda) := f(x) + \sum_{j=1}^m \lambda_j \varphi_j(x)$$

such deformations are called infinitesimally versal.

Defn A deformation  $F(x, \lambda)$  of the germ  $f(x)$  is said to be infinitesimally versal, if

$\left. \frac{\partial F}{\partial \lambda_j} \right|_{\lambda=0}$  generate the local algebra  $\mathcal{Q}_f$  under

the canonical projection  $\mathcal{O}_n \rightarrow \mathcal{Q}_f = \mathcal{O}_n / \mathcal{I}_{\mathcal{Q}_f}$ .

(Also called universal unfolding in some context).

Fact (Arnold, V.I, Mather, J)

Any infinitesimally versal deformation of a function germ is versal.

Examples

$(A^{n-1})$      $f(x) = x^n$      $\mathcal{Q}_f$  is generated by  
1,  $x$ , ...,  $x^{n-2}$

$$\therefore F(x, \lambda) = x^n + \lambda_0 + \lambda_1 x + \dots + \lambda_{n-2} x^{n-2}$$

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Fix a (mini)versal deformation  $F(x, \lambda)$  of  $f$ .  
 we obtain another family

$$V_\lambda = \{(z, \lambda) : F(z, \lambda) = 0\}$$

$\downarrow$

$\lambda$

not all fibres are nonsingular. to give this:

defn: The level bifurcation set (or - discriminant)  
 of the singularity  $f$  is (the germ) of surface  
 $\Sigma$  in the base of its miniversal deformation  
 by singular values of  $\lambda$ .

To see it's the discriminant, we go to

Example 1.  $(A_2)$   $f(x) = x^3$   $F(x, \lambda) = x^3 + \lambda_1 x + \lambda_0$ .

$$\Sigma = \{(\lambda_1, \lambda_0) \mid \begin{array}{l} F(x, \lambda) = 0 \\ \frac{d}{dx} F(x, \lambda) = 0 \end{array} \exists x\} \quad \text{(condition for having multiple roots.)}$$

$$= \{(\lambda_1, \lambda_0) \mid 27\lambda_0^2 + 4\lambda_1^3 = 0\}$$

Example 2 ( $A_3$ )  $f(x) = x^4$   $F(x, \lambda) = x^4 + \lambda_2 x^2 + \lambda_1 x + \lambda_0$

$\Sigma$  is known as a swallowtail.

Complements of Discriminants and Braid group

Eg. for  $F(x, \lambda) = x^{m+1} + \lambda_1 x^{m-1} + \dots + \lambda_{m-1} x + \lambda_m$

$$= \prod_{i=1}^{m+1} (x - x_i) \quad (\sum x_i = 0)$$

$\mathbb{C}^m \setminus \Sigma :=$  configuration space of non-ordered  $m$  distinct point

$$\pi_1(\mathbb{C}^m \setminus \Sigma) = \text{Br}(m+1) \quad \text{braid group.}$$

Admit a generalization to simple singularities.